Case Study 5: Simple ATM Deposit and Withdrawal

1. Problem Statement

Create a simple ATM program that allows a user to deposit money or withdraw money from an initial account balance.

Display the updated balance.

2. Objectives

1. Store the account balance in a variable.
2. Read transaction type and amount.
3. Use conditions to process deposits and withdrawals.
4. Display the updated balance.

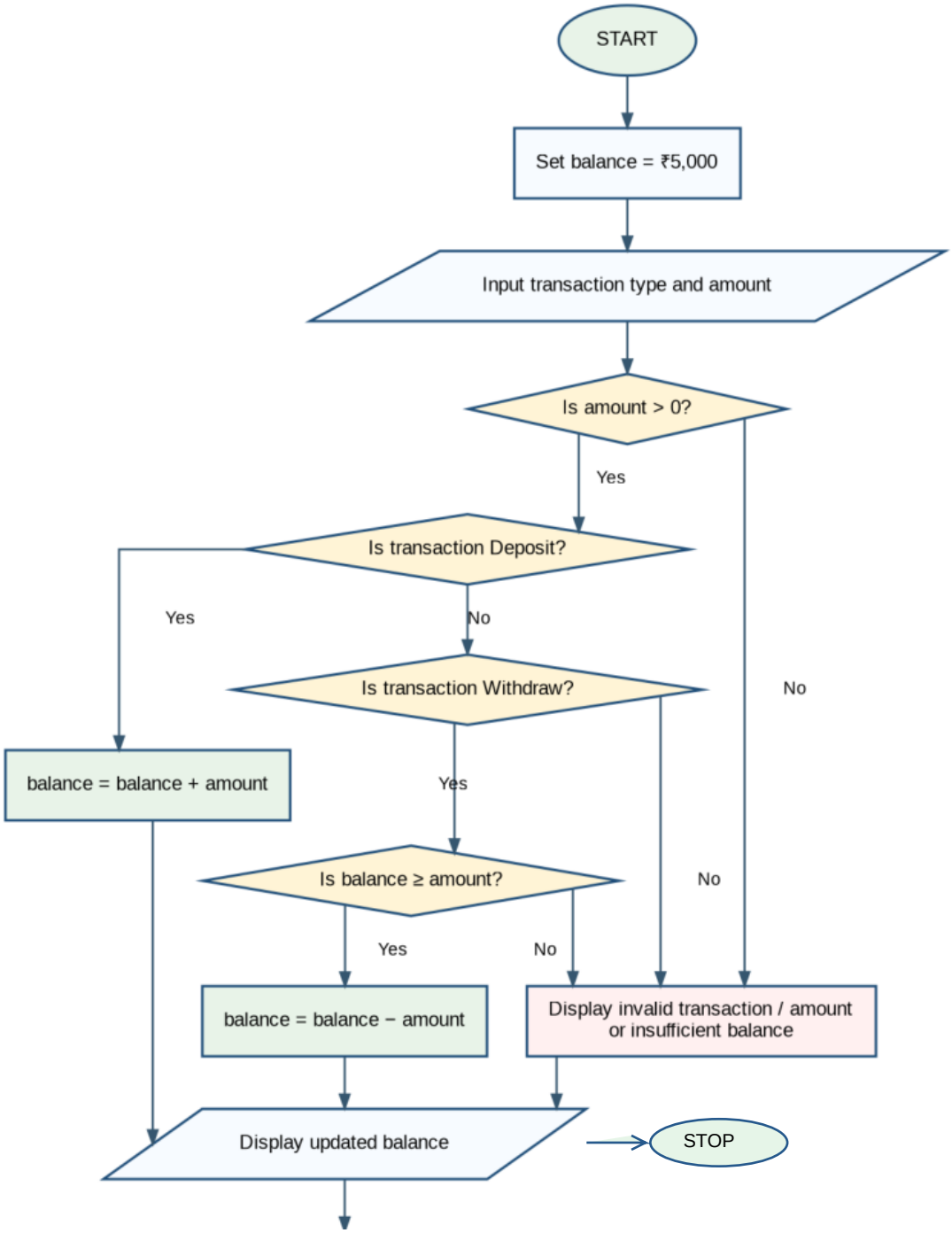
3. Assumptions

1. Initial balance is ■5,000.
2. Transaction amount must be positive.
3. Withdrawal is allowed only when sufficient balance is available.

4. Algorithm

1. Start.
2. Set balance to 5,000.
3. Read transaction type and amount.
4. If transaction is Deposit, add amount to balance.
5. If transaction is Withdraw, check balance and subtract amount if sufficient.
6. Otherwise display an invalid transaction message.
7. Display final balance.
8. Stop.

5. Flowchart



6. Pseudocode

```
balance = 5000
INPUT choice, amount
IF amount <= 0 THEN DISPLAY "Invalid amount"
ELSE IF choice = "deposit" THEN balance = balance + amount
ELSE IF choice = "withdraw" AND amount <= balance THEN
    balance = balance - amount
ELSE DISPLAY "Invalid transaction"
DISPLAY balance
```

7. Complete Python Source Code

```
balance = 5000
print("Initial balance: Rs.", balance)
choice = input("Enter deposit or withdraw: ").lower()
amount = float(input("Enter amount: Rs. "))
if amount <= 0:
    print("Invalid amount")
elif choice == "deposit":
```

```
    balance = balance + amount
elif choice == "withdraw":
    if amount <= balance:
        balance = balance - amount
    else:
        print("Insufficient balance")
else:
    print("Invalid transaction type")
print("Available balance: Rs.", balance)
```

8. Test Case and Expected Output

Initial balance = ■5,000; Choice = withdraw; Amount = ■1,200.

Updated balance = ■5,000 – ■1,200 = ■3,800.

9. Conclusion

The program simulates a basic ATM transaction using variables and conditional statements.